
Structure-Preserving Residual Initialization for Dense-to-MoE Upcycling

Visal Soeun¹ Jaehyun Sim¹ Kevin Min¹

Abstract

Mixture-of-Experts (MoE) models scale model capacity while keeping per-token compute fixed, but training them from scratch is prohibitively expensive. Upcycling initializes MoE experts from a pretrained dense model, yet the initialization strategy must balance preserving pretrained representations with encouraging expert specialization. We propose spectral residual initialization, which decomposes each dense weight matrix via singular value decomposition, preserves the top- k singular values as shared structural components, and introduces per-expert diversity only in the residual subspace. We validate on two architectures with contrasting designs: Qwen1.5 (1.8B $\rightarrow$ 14.3B MoE, with weight partitioning and a shared expert) and Mixtral (7B $\rightarrow$ 46.7B MoE, with exact dimension match and no shared expert). Spectral residual improves GSM8K accuracy over direct copy by +2.3% on Qwen1.5 and +1.2% on Mixtral, while consistently outperforming Gaussian perturbation at every matched noise scale. Analysis of expert weight dynamics shows that effective specialization depends not on how much experts differ from each other, but on which weight directions are perturbed.

1. Introduction

Mixture-of-Experts (MoE) architectures have emerged as a leading approach to scaling language model capacity without proportionally increasing inference cost (Fedus et al., 2022; Jiang et al., 2024). By routing each token to a small subset of specialized expert modules, MoE models maintain a large total parameter count while activating only a fraction per forward pass. However, training MoE models from scratch requires substantial compute and careful load-balancing, making it expensive relative to dense training.

¹Department of Electrical Engineering & Computer Science, Massachusetts Institute of Technology, Cambridge, MA 02139, USA. Correspondence to: Visal Soeun <visal@mit.edu>, Jaehyun Sim <simjay@mit.edu>, Kevin Min <km1317@mit.edu>.

Upcycling converts a pretrained dense model into an MoE architecture by using the dense weights to initialize expert modules (Komatsuzaki et al., 2023). Rather than training billions of expert parameters from random initialization, upcycling leverages the representations already learned by the dense model, potentially achieving strong performance with significantly less additional training.

The central challenge of upcycling is the *initialization dilemma*. If all experts receive identical copies of the dense weights, they begin in a symmetric state with no incentive to specialize, potentially requiring many training steps before they differentiate (Komatsuzaki et al., 2023). On the other hand, if experts are initialized with large random perturbations to break symmetry, the carefully learned dense representations may be damaged, leading to a recovery period before useful features can be re-acquired.

We propose a principled middle ground: **spectral residual initialization**.¹ For each dense weight matrix, we compute the singular value decomposition and separate it into a *structural component* (the top- k singular values and their associated directions) and a *residual component* (the remaining smaller singular values). Experts share the structural component exactly, preserving the high-energy features learned during dense pretraining, while receiving independent stochastic perturbations only in the residual subspace. This injects diversity precisely where it is least likely to damage learned representations, while still providing the symmetry-breaking signal needed for expert specialization.

We evaluate on two architectures that test spectral residual under contrasting conditions. On Qwen1.5, we convert the 1.8B dense model into the 14.3B MoE-A2.7B architecture with 60 routing experts and a shared expert. Spectral residual achieves 27.9% GSM8K test accuracy after 400 steps, a 2.3% improvement over direct copy (25.6%) and Gaussian perturbation (25.5%). On Mixtral, we convert Mistral 7B into Mixtral 8x7B with 8 experts and no shared expert, where each expert receives a full copy of the dense FFN weights, making direct copy a stronger baseline. Spectral residual achieves 26.7% after 300 steps, improving over direct copy (25.5%) by 1.2%.

¹Code is available at <https://github.com/simjay/structural-moe-upcycling>.

Our analysis reveals three findings: (1) spectral residual produces experts that are more effectively specialized and achieve better downstream accuracy than both baselines, (2) where diversity is introduced matters more than how much: Gaussian perturbation creates maximally diverse but poorly performing experts, and (3) spectrally initialized models exhibit more balanced router utilization, suggesting better use of total model capacity.

2. Related Work

Sparse Upcycling. Komatsuzaki et al. (2023) introduced the upcycling paradigm, demonstrating that copying dense model weights directly into MoE expert slots provides a strong initialization that outperforms random initialization. Their work established direct copy as the standard baseline. However, they did not explore structured perturbation strategies to encourage expert differentiation.

Expert Initialization and Diversity. Horoi et al. (2025) showed that undertraining experts (stopping early) can paradoxically improve MoE performance by preventing experts from collapsing to similar solutions. This suggests that the dynamics of expert specialization are critical to upcycling success. Our work complements this by modifying the starting point rather than the training procedure.

Instruction-Tuned Upcycling. Hui et al. (2025) extended upcycling to instruction-tuned models through parameter merging strategies that combine multiple fine-tuned checkpoints into expert initializations. Their approach requires access to multiple fine-tuned variants of the base model, whereas our method operates on a single dense checkpoint.

Scaling Laws. Liew et al. (2025) established scaling laws for upcycling, characterizing how performance improves with additional post-upcycling compute. Their results suggest that initialization quality has diminishing but non-negligible effects at large compute budgets. Our work operates in the limited-compute regime where initialization effects are most pronounced.

Partial Re-initialization. Nakamura et al. (2025) proposed Drop-Upcycling, which randomly drops and re-initializes subsets of expert weights along the intermediate dimension to promote specialization. Their method excels in long-term training (hundreds of billions of tokens) where the recovery cost of re-initialized weights is amortized. In contrast, our spectral residual method targets the limited-compute regime: rather than deleting and re-initializing features, we perturb only the magnitudes of low-energy singular values while preserving all pretrained directions. This avoids the initial performance degradation of re-initialization and provides immediate diversity benefits within a few hundred training steps.

None of the above works propose perturbing weights in a structured subspace defined by the singular value spectrum. Spectral residual initialization is complementary to these approaches and could be combined with undertraining schedules or parameter merging for further gains.

3. Background

3.1. Problem Setup

Consider a pretrained dense transformer with L layers, where each layer contains a feed-forward network (FFN) with weight matrices $W^{(\ell)} \in \mathbb{R}^{d_{\text{ff}} \times d_{\text{model}}}$ (for gate and up projections) and $W_{\text{down}}^{(\ell)} \in \mathbb{R}^{d_{\text{model}} \times d_{\text{ff}}}$ (for the down projection).

The target MoE architecture replaces each FFN with N routing experts, each with weight matrices $W_e^{(\ell)} \in \mathbb{R}^{d_{\text{expert}} \times d_{\text{model}}}$, and a learned router $g^{(\ell)}$ that selects K active experts per token. The goal of upcycling is to initialize $\{W_e^{(\ell)}\}_{e=1}^N$ from the dense weights $W^{(\ell)}$ such that the model retains pretrained capability while experts have sufficient diversity to specialize during subsequent training. When $d_{\text{expert}} < d_{\text{ff}}$, the dense FFN must be partitioned into non-overlapping chunks of size d_{expert} before replication across experts.

3.2. Existing Initialization Strategies

Direct Copy. The simplest approach copies dense FFN weights directly into each expert slot:

$$W_e = W \quad \forall e \in \{1, \dots, N\} \quad (1)$$

(or the appropriate partition when $d_{\text{expert}} \neq d_{\text{ff}}$). This perfectly preserves the pretrained function at step zero but creates a fully symmetric expert configuration. Experts must rely entirely on router gradient signal to differentiate, which can be slow.

Gaussian Perturbation. To break symmetry, each expert receives the dense weights plus independent noise:

$$W_e = W + \sigma \cdot \bar{w} \cdot \varepsilon_e, \quad \varepsilon_e \sim \mathcal{N}(0, I) \quad (2)$$

where $\bar{w} = \text{mean}(|W|)$ is a scale factor ensuring noise is proportional to weight magnitudes. While this breaks symmetry effectively, the perturbation is isotropic: it corrupts high-energy learned features just as much as low-energy residual structure.

4. Spectral Residual Initialization

Our key insight is that not all directions in weight space are equally important. The singular value decomposition of a weight matrix reveals its *spectrum*: an ordered set of singular values that separates dominant learned features (large values) from less critical residual structure (small values).

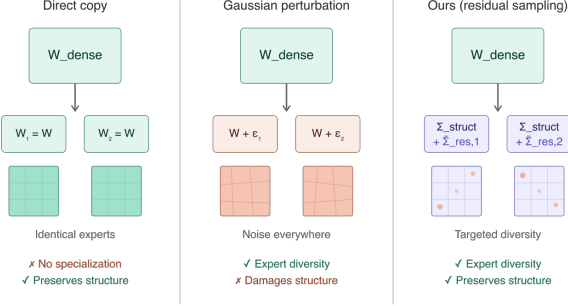


Figure 1. Overview of the three initialization strategies.

We call our approach *spectral residual initialization* because it preserves the high-energy portion of this spectrum while perturbing only the residual portion. Figure 1 illustrates the three initialization strategies. Direct copy preserves pretrained structure but creates identical experts with no specialization signal. Gaussian perturbation introduces diversity isotropically, damaging learned features along every direction. Spectral residual introduces diversity only in the residual singular value subspace, preserving structure while enabling specialization.

Step 1: Decompose. For each weight matrix W , compute the thin SVD:

$$W = U\Sigma V^\top \quad (3)$$

where $U \in \mathbb{R}^{m \times r}$, $\Sigma = \text{diag}(\sigma_1, \dots, \sigma_r)$ with $\sigma_1 \geq \dots \geq \sigma_r \geq 0$, and $V \in \mathbb{R}^{n \times r}$, with $r = \min(m, n)$.

Step 2: Separate. Split the singular values into structural and residual components:

$$\Sigma_{\text{struct}} = \text{diag}(\sigma_1, \dots, \sigma_k, 0, \dots, 0) \quad (4)$$

$$\Sigma_{\text{res}} = \text{diag}(0, \dots, 0, \sigma_{k+1}, \dots, \sigma_r) \quad (5)$$

The top- k singular values capture the dominant learned features. The remaining $r - k$ values constitute the residual.

Step 3: Perturb residual per expert. For each expert e , construct a perturbed singular value vector:

$$(\tilde{\Sigma}_e)_m = \begin{cases} \sigma_m & \text{if } m \leq k \\ \sigma_m + s \cdot \bar{\sigma}_{\text{res}} \cdot \eta_{e,m} & \text{if } m > k \end{cases} \quad (6)$$

where $\eta_{e,m} \sim \mathcal{N}(0, 1)$ are i.i.d. noise samples, $\bar{\sigma}_{\text{res}} = \text{mean}(|\sigma_{k+1}|, \dots, |\sigma_r|)$ is the mean residual singular value magnitude, and s is a scale hyperparameter controlling perturbation strength.

Step 4: Reconstruct.

$$W_e = U \cdot \text{diag}(\tilde{\Sigma}_e) \cdot V^\top \quad (7)$$

The structural singular directions (columns of U and V corresponding to the top- k values) are shared exactly across all

Algorithm 1 Spectral Residual Expert Initialization

Require: Dense weight W , num experts N , structural rank k , noise scale s

Ensure: Expert weights $\{W_1, \dots, W_N\}$

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1:  $U, \Sigma, V^\top \leftarrow \text{SVD}(W)$ 
2:  $\sigma_{\text{struct}} \leftarrow \Sigma[1:k]$  // top- $k$  singular values (preserved)
3:  $\sigma_{\text{res}} \leftarrow \Sigma[k+1:r]$  // residual singular values
4:  $\bar{\sigma}_{\text{res}} \leftarrow \text{mean}(|\sigma_{\text{res}}|)$  // scale factor
5: for  $e = 1$  to  $N$  do
6:    $\eta_e \sim \mathcal{N}(0, I) \in \mathbb{R}^{r-k}$ 
7:    $\tilde{\sigma}_e \leftarrow \text{concat}(\sigma_{\text{struct}}, \sigma_{\text{res}} + s \cdot \bar{\sigma}_{\text{res}} \cdot \eta_e)$ 
8:    $W_e \leftarrow U \cdot \text{diag}(\tilde{\sigma}_e) \cdot V^\top$ 
9: end for
10: return  $\{W_1, \dots, W_N\}$ 
    
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experts, preserving the high-energy representations. Diversity is introduced exclusively through the residual singular values, ensuring that expert differentiation occurs in directions that carry less information about the pretrained task.

This procedure has several desirable properties. The noise is scaled by the mean residual singular value, ensuring perturbations remain proportional to the existing residual structure regardless of overall weight scale. SVD is computed once per weight matrix at initialization time, costing $O(mn \cdot \min(m, n))$, which is negligible compared to training.

5. Experimental Setup

5.1. Models

We evaluate on two model families that differ in key architectural properties relevant to upcycling.

Qwen1.5. We upcycle the dense 1.8B-parameter Qwen1.5 model (Bai et al., 2023) into the MoE-A2.7B architecture (Table 1). The shared expert receives an exact copy of the full dense FFN (dimension 5504) and is not perturbed. Only the 60 routing experts (dimension 1408 each) are subject to the initialization strategy under study. Because the expert dimension is smaller than the dense FFN dimension ($1408 < 5504$), each dense weight matrix is split into 4 non-overlapping chunks along the row dimension. The 60 experts cycle through these chunks, with 15 experts assigned to each chunk. SVD is applied independently to each chunk before perturbation.

Mixtral. We upcycle Mistral 7B (Jiang et al., 2024) into the Mixtral 8x7B architecture (Table 2). Mixtral presents a contrasting test case. The expert dimension matches the dense FFN exactly ($d_{\text{expert}} = d_{\text{ff}} = 14336$), so no partitioning is needed and each expert receives a full copy of the dense weights. The absence of a shared expert means

Table 1. Qwen1.5 architecture comparison.

Property	Dense (1.8B)	MoE (A2.7B)
Total params	1.8B	14.3B
Active params	1.8B	2.7B/token
Hidden dim	2048	2048
FFN intermediate	5504	1408/expert
Layers	24	24
Routing experts	–	60
Active experts/token	–	4
Shared expert	–	1 (dim 5504)

Table 2. Mixtral architecture comparison.

Property	Dense (7B)	MoE (8x7B)
Total params	7.2B	46.7B
Active params	7.2B	12.9B/token
Hidden dim	4096	4096
FFN intermediate	14336	14336/expert
Layers	32	32
Routing experts	–	8
Active experts/token	–	2
Shared expert	–	None

routing experts bear full responsibility for processing each token, removing the compensation mechanism present in Qwen1.5. Together, these properties make direct copy a stronger baseline and expose the full impact of expert initialization quality.

5.2. Training Configuration

Both experiments train on the GSM8K training split (Cobbe et al., 2021), consisting of 7,473 grade-school math word problems with step-by-step solutions. To isolate the effect of expert initialization, we freeze all parameters except the routing expert FFN weights and the router. Attention projections (Q, K, V, O), embeddings, and layer norms remain fixed throughout training. For Qwen1.5, the shared expert is also frozen. Table 3 summarizes the training configurations. The lower learning rate for Mixtral reflects the larger model size and the stronger direct-copy baseline.

Table 3. Training configuration.

Parameter	Qwen1.5	Mixtral
Training steps	400	300
Batch size	1×4 grad. accum.	1×4 grad. accum.
Learning rate	5×10^{-4} , cosine	1×10^{-5} , cosine
Warmup	30 steps	30 steps
Optimizer	AdamW (8-bit)	AdamW (8-bit)
Precision	bfloat16	bfloat16
Seq. length	2048	2048
Grad. ckpt.	Enabled	Enabled
Hardware	$1 \times$ A100-80GB	$8 \times$ A100-80GB

5.3. Evaluation

We evaluate on the full GSM8K test split (1,319 problems) using 4-shot prompting with greedy decoding. The model generates step-by-step solutions, and we extract the final numeric answer (following the “####” delimiter) for comparison with ground truth. For Qwen1.5, we track expert specialization throughout training via mean pairwise cosine similarity and normalized L2 distance between expert weight vectors within the same partition group.

Our evaluation uses few-shot prompting without instruction tuning, which results in lower absolute accuracy than instruction-tuned baselines on public leaderboards. Since all configurations are evaluated under identical conditions, we focus on relative comparisons between initialization strategies. For Qwen1.5, the dense baseline (Qwen1.5-1.8B without upcycling) achieves 33.9% under our protocol, establishing an upper reference for preserved pretrained capability.

5.4. Configurations

For Qwen1.5, we test five configurations: direct copy, Gaussian perturbation with $\sigma \in \{0.5, 1.0\}$, and spectral residual with $k = 512$ and scale $s \in \{0.1, 0.5\}$. Here $k = 512$ preserves the top 512 out of 1408 singular values as structural, perturbing the remaining 896 residual values.

For Mixtral, we test five configurations: direct copy, Gaussian perturbation with $\sigma \in \{0.1, 0.5\}$, and spectral residual with $k = 512$ and scale $s \in \{0.1, 0.5\}$. Here $k = 512$ preserves the top 512 out of 4096 singular values, perturbing the remaining 3584. The Gaussian baselines use smaller sigma values (0.1, 0.5) than those tested for Qwen1.5 (0.5, 1.0), as the exact dimension match makes the Mixtral baseline more sensitive to perturbation.

6. Results

6.1. Main Results

Qwen1.5. Table 4 presents GSM8K test accuracy and expert divergence metrics across all initialization strategies after 400 training steps. Spectral residual with scale 0.5 achieves the highest accuracy among all Qwen1.5 configurations, improving over direct copy by 2.3%. Both spectral residual configurations outperform the baselines, while Gaussian perturbation fails to improve over direct copy regardless of noise magnitude, despite creating more diverse experts. The gap between the dense baseline (33.9%) and the best upcycled model (27.9%) reflects the overhead of introducing routing with only 400 steps of training.

Mixtral. Table 5 presents GSM8K test accuracy for the Mixtral experiment after 300 training steps. Spectral residual with $s=0.1$ achieves the best accuracy at 26.7%, improving

Table 4. Qwen1.5 results on GSM8K test set (1,319 problems). Cosine similarity measures mean pairwise similarity between experts (lower = more specialized).

Method	Acc.	Δ	Final Cos Sim
Dense baseline	33.9%	—	—
Spectral $k=512, s=0.5$	27.9%	+2.3	0.862
Spectral $k=512, s=0.1$	27.1%	+1.5	0.902
Direct copy	25.6%	0.0	0.903
Gaussian $\sigma=0.5$	25.5%	-0.1	0.797
Gaussian $\sigma=1.0$	25.5%	-0.1	0.586

Table 5. Mixtral results on GSM8K test set (1,319 problems).

Method	Acc.	Δ
Spectral $k=512, s=0.1$	26.7%	+1.2
Gaussian $\sigma=0.1$	26.1%	+0.6
Direct copy	25.5%	0.0
Spectral $k=512, s=0.5$	18.7%	-6.8
Gaussian $\sigma=0.5$	18.1%	-7.3

over direct copy by 1.2%. At each noise scale, spectral residual outperforms Gaussian: 26.7% vs 26.1% at $s=0.1$, and 18.7% vs 18.1% at $s=0.5$. The $s=0.5$ variants suffer severe degradation for both methods, losing 6–7% relative to direct copy. This sensitivity is expected: because Mixtral’s expert dimension matches the dense FFN exactly, direct copy preserves the full pretrained representation, and large perturbations destroy too much of it to recover in 300 steps.

6.2. Expert Behavior

Qwen1.5. Figure 2 shows the evolution of expert cosine similarity over 400 training steps (lower values indicate greater expert specialization). The curves reveal qualitatively different specialization dynamics depending on initialization. Direct copy starts at approximately 1.000 and decreases slowly to 0.903 by step 400, remaining trapped in a near-symmetric basin. Gaussian $\sigma=1.0$ starts at 0.623 but barely changes during training (final: 0.586), as the unstructured diversity provides little room for productive gradient-driven adaptation. Spectral residual $s=0.5$ starts at 0.949 and decreases smoothly to 0.862, showing that the structured initial differences between experts allow training to push them further apart over time.

The L2 divergence metric (Figure 3) tells a complementary story: higher values indicate greater divergence between expert weight vectors. Direct copy starts at zero and grows to 0.441. Spectral $s=0.5$ starts at 0.320 and grows to 0.526. Gaussian $s=1.0$ starts at 0.868 and grows only to 0.910. The critical observation is that initial diversity does not predict final task performance: Gaussian $s=1.0$ is most diverse throughout training yet performs worst.

Beyond weight-space divergence, we examine router uti-

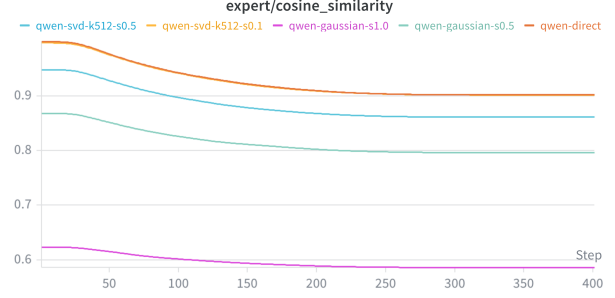


Figure 2. Mean pairwise cosine similarity between expert weights over training (Qwen1.5).

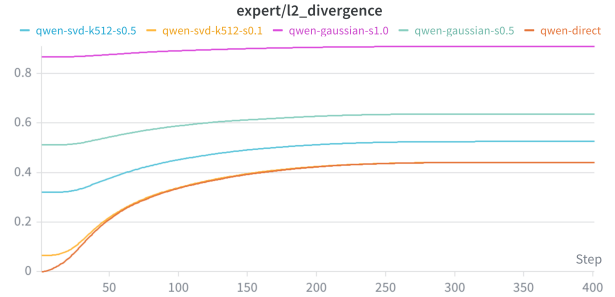


Figure 3. Normalized L2 divergence between expert weights over training (Qwen1.5).

lization (Figure 4). Spectrally initialized models exhibit a tighter interquartile range in expert usage fractions, indicating more balanced token routing across experts. Under direct copy, the router has no informative signal for differentiation, leading to arbitrary routing patterns. Under Gaussian perturbation, uneven damage across experts can cause the router to favor the least-corrupted subset. Spectral residual provides each expert with a distinct but structurally sound starting point, giving the router meaningful differences to exploit from step zero.

Expert divergence metrics were not logged during Mixtral

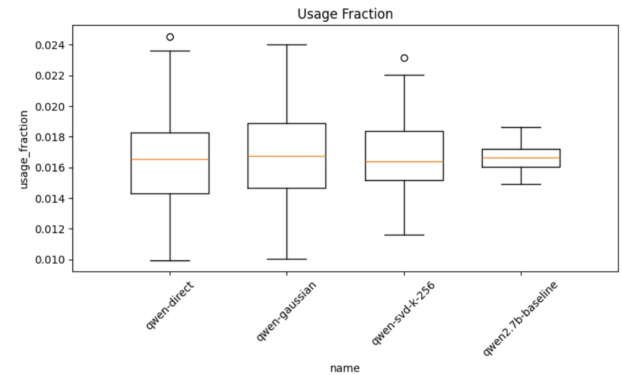


Figure 4. Expert usage fraction distributions (Qwen1.5).

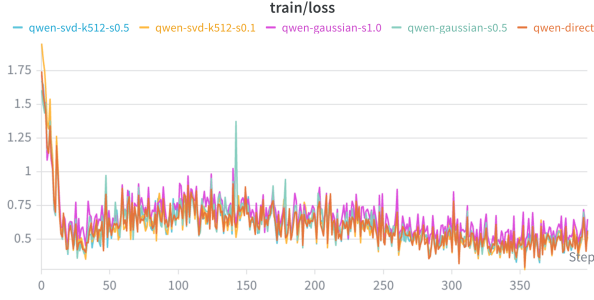


Figure 5. Training loss over 400 steps (Qwen1.5).

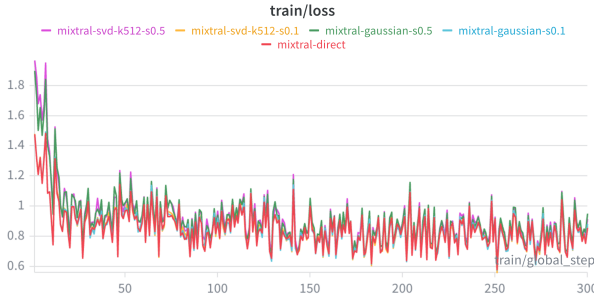


Figure 6. Training loss over 300 steps (Mixtral).

training runs, so this analysis is limited to Qwen1.5.

6.3. Training Dynamics

Qwen1.5. Figure 5 shows training loss curves for all Qwen1.5 configurations. All methods converge to similar loss values (approximately 0.5) by step 400, with no clearly separable trajectories after the initial transient (steps 0 to 50). This convergence is expected for two reasons. First, the limited training budget constrains how much separation can emerge. Second, the shared expert processes every token and provides a strong baseline signal regardless of routing expert initialization, absorbing much of the training loss.

Mixtral. The Mixtral training loss curves (Figure 6) show a clearer separation in the early phase. The $s=0.5$ variants (both spectral residual and Gaussian) start at a loss of approximately 1.9–2.0, while $s=0.1$ variants and direct copy all start near 1.48. This confirms that larger perturbations cause measurable initial degradation. By step 300, all methods converge to similar final loss values (0.8–0.9). Unlike Qwen1.5, this convergence cannot be attributed to a shared expert absorbing the loss, since Mixtral has none. Instead, the optimizer compensates for different initializations within the training budget. Despite similar final loss, downstream accuracy still differs, reinforcing the finding from Qwen1.5 that training loss alone does not capture the quality of expert specialization.

In both experiments, downstream GSM8K accuracy diverges meaningfully despite similar training loss. This confirms that training loss is not a sufficient proxy for the quality of expert specialization.

6.4. Analysis

Our Qwen1.5 results reveal a non-monotone relationship between expert diversity and downstream performance. If diversity alone were beneficial, Gaussian $\sigma=1.0$ should perform best, as it produces the lowest cosine similarity (0.586). Instead, it performs no better than direct copy (25.5% vs 25.6%). Spectral $s=0.5$ achieves the highest accuracy (27.9%) at a moderate similarity of 0.862. This reveals that the *subspace* of diversity matters critically. Gaussian noise perturbs all singular directions equally, damaging high-energy features that encode useful pretrained representations. Spectral residual perturbs only low-energy residual directions, creating diversity where perturbation is safe while preserving dominant features. The best performance comes not from maximum diversity, but from perturbing only the less important weight directions enough to break symmetry without damaging the important ones.

Cross-model comparison. The two experiments reveal that the optimal perturbation scale differs between architectures: $s=0.5$ works best for Qwen1.5, while $s=0.1$ works best for Mixtral. Several factors may explain this difference. Qwen1.5 experts receive only a chunk of the dense FFN (1408 out of 5504 dimensions), so each expert’s starting point already loses information and can tolerate more perturbation. In contrast, Mixtral experts receive the full dense FFN (14336 dimensions), preserving more pretrained structure that is sensitive to disruption. The larger model size of Mixtral (7B vs. 1.8B) may also contribute, as larger pretrained weights encode more information per direction. With only two architectures, we cannot isolate which factor is primary. Across both architectures and all matched noise scales, spectral residual consistently outperforms Gaussian perturbation.

7. Discussion

Reliability of results. All experiments use single training runs, so we cannot report confidence intervals across random seeds. However, several aspects of the experimental design strengthen confidence in the findings. First, spectral residual outperforms both baselines on both architectures, providing cross-architecture replication of the core result. Second, at every matched noise scale where we can directly compare spectral residual and Gaussian perturbation (four comparisons: two scales on each architecture), spectral residual outperforms Gaussian. Third, our evaluation uses the full GSM8K test set (1,319 problems), reducing variance in the accuracy estimates. Finally, the expert divergence met-

rics on Qwen1.5 provide a mechanistic explanation for the accuracy differences: spectral residual experts specialize more effectively during training, which is consistent with the downstream accuracy gains rather than being a statistical artifact.

Generalizability. Our experiments are limited to a single task (math reasoning on GSM8K) and two model families. The benefit of spectral residual initialization is likely to be most pronounced in settings where expert specialization matters for downstream performance, such as tasks requiring diverse reasoning strategies. For tasks that do not require different experts to handle different types of inputs, the advantage of structured diversity may be smaller. Similarly, our limited-compute regime (300–400 steps) maximizes the effect of initialization quality. At substantially larger compute budgets, training may eventually overcome poor initialization, reducing the gap between methods (Liew et al., 2025). We note that both architectures tested here use SwiGLU-based FFN experts. Whether the same spectral approach extends to other expert formulations (e.g., standard MLP experts or attention-based experts) remains an open question.

Practical guidelines. Based on our two experiments, larger models with exact dimension matches between dense and expert weights (as in Mixtral) benefit from conservative perturbation scales ($s \leq 0.1$), while smaller models where experts receive partitioned weight chunks (as in Qwen1.5) can tolerate larger scales ($s \approx 0.5$). In both cases, spectral residual with an appropriately chosen scale outperforms both direct copy and Gaussian perturbation with matched noise magnitude.

Choice of k . We use $k = 512$ for both architectures, which corresponds to different fractions of the total singular values: 512 out of 1408 (36%) for Qwen1.5 expert chunks, and 512 out of 4096 (12.5%) for Mixtral. The fact that the same absolute k works well across both settings suggests some robustness to this hyperparameter, though we did not systematically sweep over k values. Preliminary experiments with $k < 512$ on Mixtral produced poor results, consistent with the finding that preserving fewer structural directions exposes more of the pretrained representation to perturbation. A principled selection criterion, such as choosing k to preserve a target fraction of the spectral energy for each weight matrix, is a natural direction for future work.

Compute constraints. Our experimental scope reflects limited GPU access: a single A100-80GB for Qwen1.5 and eight A100-80GB GPUs for Mixtral. This constrained the training budget to 300–400 steps, which is why we evaluate on a single benchmark (GSM8K) rather than a multi-task suite, and why we report single-run results rather than averaging over multiple random seeds. The short training regime is further justified by our observation that extending

Qwen1.5 training to 2,000 steps led to overfitting without additional accuracy gains. Within these constraints, we prioritized breadth of architectural coverage (two contrasting MoE designs) over depth of evaluation on any single model, as cross-architecture validation provides stronger evidence for the method’s generality than repeated runs on a single architecture.

8. Conclusion

We have presented spectral residual initialization, a simple and principled strategy for dense-to-MoE upcycling that decomposes pretrained weight matrices into structural and residual components, then introduces per-expert diversity exclusively in the residual subspace. Across two architectures with contrasting designs, spectral residual consistently outperforms both direct copy and Gaussian perturbation: +2.3% on Qwen1.5 (1.8B to 14.3B MoE, 400 steps) and +1.2% on Mixtral (7B to 46.7B MoE, 300 steps). Spectrally initialized experts also exhibit more balanced router utilization, suggesting better use of total model capacity.

Our analysis reveals that expert initialization quality depends not on how different the experts are from each other, but on which weight directions are perturbed. Unstructured Gaussian noise creates maximally diverse but poorly performing experts by damaging pretrained features. Spectral residual confines diversity to the low-energy residual subspace, providing a productive starting point for expert specialization while preserving the dominant representations that make upcycling effective in the first place. The Mixtral experiment confirms that this benefit is not an artifact of Qwen1.5’s shared expert or dimension partitioning, though the optimal perturbation scale differs between architectures ($s=0.5$ for Qwen1.5, $s=0.1$ for Mixtral).

Limitations. We evaluate on a single benchmark (GSM8K) at two model scales with a fixed structural rank ($k = 512$). All results are from single training runs without confidence intervals. The optimal k likely depends on the spectral properties of each weight matrix, and we did not sweep over this hyperparameter. With only 300–400 steps of training, initialization effects are maximally pronounced and may diminish at larger compute budgets (Liew et al., 2025). Expert divergence metrics are reported only for Qwen1.5, as they were not logged during Mixtral training.

Future work. Several directions would strengthen and extend these findings. First, evaluating on additional benchmarks (e.g., code generation, general language understanding, summarization) would test whether spectral residual’s benefits transfer beyond math reasoning. Second, running each configuration with multiple random seeds would establish confidence intervals on the accuracy differences and confirm statistical significance. Third, extending training

to thousands of steps would reveal whether spectral residual’s advantage persists, shrinks, or grows as training progresses, connecting to the scaling laws established by Liew et al. (2025). Fourth, a systematic sweep over the structural rank k (e.g., $k \in \{128, 256, 512, 1024\}$) would characterize sensitivity to this hyperparameter and potentially yield a spectral-energy-based selection criterion that adapts k per layer. Fifth, the current method perturbs only singular values; perturbing the singular vectors U and V as well could provide richer diversity at the cost of more disruption to pre-trained directions. Finally, combining spectral residual with complementary techniques such as undertraining schedules (Horoi et al., 2025) or parameter merging (Hui et al., 2025) may yield compounding benefits.

Impact Statement

This work aims to reduce the computational cost of training Mixture-of-Experts language models by improving how experts are initialized during upcycling. If effective at scale, better initialization strategies could allow practitioners to obtain high-quality MoE models with less post-upcycling training, reducing GPU hours and the associated energy consumption. This is particularly relevant as MoE architectures grow to hundreds of billions of parameters, where even modest reductions in training steps translate to meaningful resource savings. Our method operates at initialization time and does not change the model’s capabilities or behavior at deployment, so it does not introduce new risks beyond those already present in the underlying dense and MoE models.

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